

Rakesh Narla

DevOps Engineer

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Summary

DevOps Engineer with 6+ years of experience in building, automating, and maintaining cloud-based infrastructure. Skilled in **Terraform, Kubernetes, CI/CD pipelines, cloud platforms** (AWS, GCP), and **automation tools** (Ansible, Python). Known for collaborating in teams to streamline infrastructure management, optimize processes, and ensure the security and scalability of critical systems. Adept at solving problems with cloud technologies, monitoring, and infrastructure automation.

Professional Experience

Senior DevOps Engineer

Motherson Technologies Services Limited | Bosch Project

📅 03/2025 - Present 📍 Bangalore, India

Contributed to the development and modernization of enterprise DevOps and cloud platform capabilities for a large-scale Bosch environment, supporting CI/CD, multi-cloud infrastructure automation, cloud governance, and secure application delivery across GCP, AWS, and Azure using Jenkins, GitHub Actions, Terraform, Kubernetes, and Helm.

- **Jenkins Shared Library & CI/CD Enhancements:** Enhanced existing **Jenkins Shared Library** capabilities by extending reusable Groovy pipelines, deployment components, and CI/CD integrations across multiple technology stacks and deployment environments.
- **Jenkins to GitHub Actions Migration:** Contributed to the modernization of CI/CD by migrating existing **Jenkins-based workflows to GitHub Actions**, developing reusable workflows and adapting existing build, test, deployment, and release processes.
- **CI/CD Technology Modernization:** Introduced and integrated newer technologies such as **Playwright, Bruno**, and other testing and automation tools into existing CI/CD workflows, expanding automated testing and improving developer experience.
- **Multi-Platform Deployment Automation:** Extended existing deployment capabilities across **Kubernetes, Helm, Azure AKS, AWS ECS, OpenShift**, and other platforms, enabling consistent application delivery across multiple environments.
- **DevSecOps & Pipeline Security:** Enhanced CI/CD security capabilities by integrating **Checkmarx, JFrog Xray, GitLeaks, OWASP ZAP, and SBOM generation**, strengthening automated security and software supply-chain validation.
- **Cloud Platform Engineering:** Worked across **GCP and AWS** supporting **Terraform-based infrastructure provisioning, IAM, cloud governance, project/account lifecycle management, and platform automation**.
- **GCP Identity & Governance:** Implemented **GitHub OIDC with GCP Workload Identity Federation** for keyless Terraform authentication and contributed to **GCP IAM, project lifecycle, access control, tagging, observability, and provisioning** standards.
- **Cloud Platform Enablement:** Supported **Databricks and multi-cloud platform initiatives**, including infrastructure automation, self-service provisioning, IAM standardization, and cloud cost optimization.

Key Contributions:

- **CI/CD Modernization:** Led the migration of enterprise CI/CD capabilities from **Jenkins Shared Libraries to GitHub Actions**, developing reusable workflows and standardizing deployment patterns across application teams.
- **Jenkins Platform Engineering:** Built and enhanced reusable **Jenkins Shared Library** components and deployment abstractions across **Kubernetes/Helm, AWS ECS, Azure AKS, OpenShift**, and other delivery platforms, reducing pipeline duplication and improving deployment consistency.
- **Cloud Platform Enablement:** Supported **CET** initiatives across **GCP and AWS**, automating cloud provisioning and governance with **Terraform and GitHub Actions**, while implementing IAM, project/account lifecycle, and self-service platform capabilities.
- **Cloud Security & Governance:** Strengthened enterprise cloud delivery through **GCP Workload Identity Federation/OIDC, IAM standardization, security controls, and governance practices**, enabling secure, scalable infrastructure automation.

Associate Infrastructure L2

Publicis Sapient | HSBC project

📅 03/2022 - 03/2025 📍 Bangalore, India

Contributed to the setup, configuration, and maintenance of Kafka and Zookeeper clusters for a large-scale banking application, supporting real-time data processing and event-streaming capabilities.

- **Kafka & Zookeeper Implementation:** Worked alongside team members to deploy **Kafka** and **Zookeeper** for real-time messaging and data streaming.
- **Cloud Security & VPC Setup:** Contributed to configuring **Virtual Private Cloud (VPC)** networks for secure, isolated environments. Assisted in implementing **firewall rules** to control inbound and outbound traffic, enhancing the security posture of the cloud infrastructure.
- **Secrets Management with Vault:** Collaborated with team members to implement **HashiCorp Vault** for secure management of API keys, secrets, and credentials. Helped ensure compliance with security policies by automating the management and rotation of sensitive data.
- **Infrastructure as Code (IaC):** Designed and deployed infrastructure using **Terraform**, enabling consistent and repeatable provisioning of cloud resources, reducing deployment times and errors by 40%.
- **Network Automation with Terraform:** Automated the creation of GCP networking components, including **VPCs, subnets, firewalls, and routes** using **Terraform**.
- **CI/CD Pipeline Automation:** Assisted in the setup and optimization of **CI/CD pipelines** using **Jenkins** and **GitLab**, automating build, test, and deployment processes. This resulted in faster and more reliable deployments with reduced manual errors.
- **Designed and implemented CI/CD pipelines using GitHub Actions** to automate the deployment of microservices to GKE. Integrated with Docker for containerization and Helm for managing Kubernetes manifests, significantly reducing deployment time and improving release reliability.
- **Service Mesh with Istio:** Assisted in deploying and configuring **Istio** as a **service mesh** to manage microservices communication, providing features such as **traffic management, security, and observability**. Implemented **mTLS** for secure service-to-service communication, ensuring encrypted data flow between microservices.
- **Monitoring & Alerts:** Supported the integration of **Prometheus** and **Grafana** to monitor system performance and set up dashboards. Helped establish proactive **alerting systems** like **Splunk** and **SplunkForwarder** to identify and address issues in real-time, minimizing downtime.
- **Security & Compliance:** Ensured compliance with **banking regulations** by implementing encryption, access control, and audit logging for Kafka topics, securing sensitive data in transit and at rest.

Key Contributions:

- Enhanced data streaming capabilities with **Kafka** for real-time transaction processing.
- Increased system security and compliance by implementing **Vault** for secrets management.
- Automated deployments and improved release velocity through optimized **CI/CD pipelines**.
- Automated infrastructure provisioning and configuration management using **Ansible Tower**, enabling centralized control and streamlined workflows.
- Developed and optimized **Ansible Playbooks** for efficient deployment, patch management, and application configuration across multiple environments.

DevOps Engineer

Tata Consultancy Services

📅 02/2020 - 03/2022 📍 Bangalore, India

Managed end-to-end DevOps processes including infrastructure automation, continuous integration/deployment, and performance monitoring for global enterprise clients.

- **Cloud Infrastructure Deployment:** Worked on **GCP** cloud platform to design and deploy infrastructure solutions, leading to a **15%** improvement in application uptime and stability. Creating the required firewall rules and routes with proper network tags for secured communication using **VPC**
- **Automation & Scripting:** Automated repetitive tasks using **Python, Bash** and **Groovy** reducing manual effort and error rates by **50%**.
- **CI/CD Implementation:** Assisted in building and optimizing **CI/CD pipelines** using **Jenkins** and **GitLab**, accelerating software release cycles and enhancing deployment efficiency
- **On-Premise to Cloud Migration:** Contributed to a large-scale **on-premise to cloud migration** project, helping transition legacy systems to **Google Cloud Platform** by automating deployment pipelines and ensuring minimal downtime during cutover.
- **Cluster Maintenance & Upgrades:** Managed **Kubernetes(GKE) clusters** and **Docker containers**, performing regular maintenance tasks, patching, and upgrading clusters to ensure reliability and optimal performance.
- **Monitoring & Alerting:** Set up **Splunk** logging and **GCP alerting** to ensure system health, enabling proactive resolution of issues before customer-facing impact.

Key Contributions:

- Migrated 50+ applications to **GCP**, improving scalability and reducing operational costs.
- Automated infrastructure and deployment processes using **Ansible**, cutting down manual tasks and minimizing errors.
- Enhanced security and compliance with automated **Venafi** certificate management and **Vault** for secrets storage.

Skills & Technologies

- **Cloud Platforms:** GCP, AWS
- **Infrastructure as Code (IaC):** Terraform, CloudFormation
- **Automation Tools:** Ansible, Python, Bash, Jenkins
- **Containerization & Orchestration:** Docker, Kubernetes, Helm
- **CI/CD Tools:** GitLab, Jenkins, ArgoCD
- **Monitoring & Logging:** Prometheus, Grafana, ElasticSearch, Splunk
- **Service Mesh:** Istio
- **Secrets Management:** HashiCorp Vault
- **Message Brokers:** Kafka, Zookeeper, Kafdrop
- **Certificate Management:** Venafi, SSL/TLS, Google secrets Manager
- **Security & Compliance:** Identity and Access Management (IAM)

Project Highlights

Motherson Technologies – Bosch | Enterprise Cloud & DevOps Platform

- Engineered and modernized **CI/CD platforms** using **Jenkins Shared Libraries** and **GitHub Actions**, including reusable pipeline development and migration of delivery workflows from Jenkins to GHA.
- Contributed to **CET cloud platform engineering** across **GCP and AWS** using **Terraform** and **GitHub Actions**, covering cloud provisioning, IAM, governance, and standardized platform operations.

- Implemented **GCP Workload Identity Federation with GitHub OIDC** and reusable **Kubernetes/Helm deployment automation**, enabling secure cloud authentication and consistent multi-environment application delivery.

Publicis Sapient - Kafka & Zookeeper for Banking Application

- Deployed **Kafka** clusters to support real-time transaction data processing.
- Worked as part of a team to ensure **Zookeeper** provided high availability and fault tolerance in mission-critical systems.
- Worked on deploying **Istio** for microservices communication, enabling **traffic management**, **security policies**, and **observability**.

Publicis Sapient - Automated CI/CD Pipeline Implementation

- Supported the implementation of **CI/CD pipelines** using **GitLab**, automating deployments and enhancing deployment speed.

TCS - Cloud Migration from On-Premise to GCP

- Contributed to migrating legacy applications to **GCP**, improving scalability and reducing infrastructure costs.
- Assisted with infrastructure provisioning and management using **Terraform**.
- Assisted in managing **Kubernetes(k8s) clusters** for hosting containerized applications, optimizing resource allocation and ensuring high availability.

Main Achievements

- **Enterprise DevOps & CI/CD Modernization:** Engineered reusable **Jenkins Shared Libraries** and **GitHub Actions** workflows, including modernization of Jenkins-based delivery to GHA and standardized CI/CD automation across technology stacks.
- **Multi-Cloud Platform Engineering:** Engineered cloud automation across **GCP and AWS** using **Terraform and GitHub Actions**, supporting provisioning, IAM, governance, and standardized platform operations within the **Central Engineering Team (CET)**.
- **Cloud Security & Identity:** Implemented **GCP Workload Identity Federation with GitHub OIDC**, enabling keyless authentication for Terraform workflows and eliminating long-lived service-account credentials from CI/CD pipelines.
- **Cloud Migration & Infrastructure Automation:** Contributed to the migration of **50+ applications from on-premise to GCP**, using **Terraform and Kubernetes** to automate infrastructure provisioning and support scalable, highly available workloads.
- **Kubernetes & Cloud-Native Engineering:** Managed and automated **Kubernetes/GKE** environments using **Helm**, with experience in **Istio service mesh, mTLS, traffic management, and observability** for production workloads.
- **Event-Driven Platform Engineering:** Contributed to deploying **Kafka and Zookeeper** for a large-scale banking application, supporting real-time transaction processing, high availability, and event-driven workloads.
- **DevSecOps & Secrets Management:** Integrated enterprise security controls including **Checkmarx, JFrog Xray, GitLeaks, OWASP ZAP, SBOM generation, HashiCorp Vault, and IAM/RBAC** into cloud and CI/CD environments.

Certifications

- Certified Kubernetes Administrator (**CKA**)
- Certified Kubernetes Security Specialist (**CKS**)
- Google Cloud Associate (**GCP**)

Education

Bachelor of Technology (B.Tech)

YSR Engineering College of Yogi Vemana University

GPA - 6.1